

# Convergence to Lévy marginals for a re-scaled rough Heston model with jumps

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## Abstract

We prove weak convergence of the log-price marginals for a re-scaled rough Heston model to the marginals of a Normal Inverse Gaussian (NIG) Lévy process. This establishes a direct connection between the two classes of processes, without using a Markovian approximation as in the reversionary Heston model of [AC24], where the true Hurst index  $H = \frac{1}{2}$ , or limits of hyper-rough models as in [AAR25], where  $H \searrow -\frac{1}{2}$ . We then extend these results to the case of the rough Hawkes–Heston model introduced in [BPS24], where the variance process has an additional self-exciting jump component, and the log-price process has a leverage/feedback term driven by the same component which generates negative jumps/skew. We also provide an explicit characterization of the limiting subordinated Lévy process with drift, which extends the NIG class.

## 1 Introduction

For a given  $\alpha \in (\frac{1}{2}, 1)$ , we consider stochastic processes  $X = (X_t)_{t \geq 0}$  and  $V = (V_t)_{t \geq 0}$  with dynamics

$$dX_t = -\left(\frac{1}{2} + \int_{\mathbb{R}_+} (e^{-\Lambda z} - 1 + \Lambda z) \nu(dz)\right) V_t dt + \sqrt{V_t} dB_t - \Lambda d\tilde{J}_t, \quad X_0 = 0, \quad (1)$$

$$V_t = V_0 + \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} \left( \lambda(\theta - V_s) ds + \sigma \sqrt{V_s} dW_s + d\tilde{J}_s \right). \quad (2)$$

Here,  $B = (B_t)_{t \geq 0}$  and  $W = (W_t)_{t \geq 0}$  are two standard Brownian motions with correlation parameter  $\rho \in (-1, 0]$ , that is,  $\langle B, W \rangle_t = \rho t$  for all  $t \geq 0$ . The process  $\tilde{J} = (\tilde{J}_t)_{t \geq 0}$  is the purely discontinuous local martingale defined by

$$\tilde{J}_t = \int_0^t \int_{\mathbb{R}_+} x (N(dx, ds) - V_s \nu(dx) ds), \quad t \geq 0,$$

where  $N(dx, dt)$  is an integer-valued random measure with compensator  $V_t \nu(dx) dt$ , and  $\nu$  is a non-negative measure with support contained in  $(0, \infty)$  and finite second moment, namely

$$\nu((-\infty, 0]) = 0, \quad \int_{\mathbb{R}_+} x^2 \nu(dx) < \infty.$$

Since the compensator of  $N$  is proportional to the solution process  $V$ , the model exhibits a self-exciting jump mechanism.

Equation (2) is an affine stochastic Volterra equation (SVE) of convolution type with jumps, whose weak well-posedness is established in [AJ21, BLP24]. We also refer to [ACL21] for the analysis of more general dynamics beyond the affine setting. Since the fractional kernel  $t \mapsto \frac{1}{\Gamma(\alpha)} t^{\alpha-1}$  is singular and introduces a dependence on the past,  $V$  is neither a semimartingale nor Markovian. On the other hand, we notice that the process  $X$  in (1) is fully determined by  $V$ , the measure  $\nu$ , and the leverage parameter  $\Lambda > 0$ .

The pair  $(X, V)$  constitutes the rough Hawkes–Heston stochastic volatility model introduced in [BPS24], where  $X$  represents the log-price process and  $V$  the variance process. The model can be viewed as an extension of the rough Heston model of [JR16] incorporating the jump component  $\tilde{J}$ . This term admits only positive jumps and drives both the variance process  $V$ , see (2), and the log-price process  $X$  in (1). Owing to the leverage coefficient  $-\Lambda$ , the corresponding jumps in  $X$  are negative. Because of this interpretation, we shall henceforth refer to  $X$  as the log-stock-price process and to  $V$  as the variance process.

As in the continuous case  $\nu = 0$ , the model (1)-(2) belongs to the class of affine processes. As a consequence, [BLP24] derives a semi-closed-form expression for the conditional Fourier–Laplace transform of  $X$  in terms of deterministic nonlinear Volterra integral equations (VIEs). From an applied perspective, this property is particularly important, as it allows for the pricing of options on the underlying asset via Fourier inversion techniques, thereby avoiding more computationally demanding methods such as Monte Carlo simulation.

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In this paper, we leverage this formula to investigate weak scaling limits of the time- $t$  marginal distribution of  $X$ , for every  $t > 0$ . More specifically, we show that a suitably rescaled version of  $X$  in (1), indexed by a parameter  $\varepsilon$  and denoted by  $X^\varepsilon$ , converges in distribution to a subordinated Lévy process with drift belonging to a class that extends the *Normal Inverse Gaussian* (NIG) family. The main technical tool is an asymptotic result for deterministic VIEs due to [FGS21], which we exploit to characterize the limit of the moment generating function (mgf) of  $X^\varepsilon$ . The weak convergence is then established by identifying a drifted time-changed Lévy process whose mgf coincides with this limiting expression.

This work fits into a recent strand of the literature investigating connections between Volterra–Heston-type models and Lévy processes. In the continuous setting, corresponding to  $\nu = 0$ , the first paper to identify such a connection was [AC24]. There, the authors consider dynamics driven by a shifted power-law kernel  $t \mapsto (t + \varepsilon)^{H - \frac{1}{2}}$  and, through a suitable Markovian approximation called the reversionary Heston model, show that for  $H = -\frac{1}{2}$  the log-price process [resp., the integrated variance] converges weakly to an NIG process [resp., an *Inverse Gaussian* (IG) process] as the shift parameter  $\varepsilon \rightarrow 0$ . This approach therefore establishes an indirect connection based on proxy models with Hurst index  $H = \frac{1}{2}$ , since the dynamics remain driven by a standard Brownian motion. A direct connection is established in [AAR25], where IG processes arise as weak limits of the integrated variance in so-called hyper-rough Heston models. These models are driven by power-law kernels that are only locally integrable, namely  $t \mapsto t^{H - \frac{1}{2}}$ , with Hurst index  $H = \alpha - \frac{1}{2} \in (-\frac{1}{2}, 0)$ . In this regime, IG processes emerge in the limit  $H \searrow -\frac{1}{2}$ . We also refer the reader to the recent work [AAS26], which studies nonlinear processes – called *Volterra clocks* – and characterizes their weak limits as first-passage times of Brownian motion to curved boundaries. In the jump setting, namely when  $\nu \neq 0$ , weak scaling limits of the integrated variance process  $t \mapsto \int_0^t V_s ds$ , where  $V$  is given by (2), are investigated in [BF25]. There, the limiting Lévy process is characterized as first-passage time to negative barriers of a spectrally positive Lévy process.

We now review some interesting (related) works with similar dynamics to (1)–(2), albeit with different modelling objectives. In the no-jumps case, Sections 3.1–3.2 in [BJ26] prove that the  $V$  process in (2) for the rough Heston (rHeston) model has a positive atom at zero. Specifically, using the solution  $y_\lambda(\cdot)$  to the VIE associated with  $\mathbb{E}[e^{-\lambda V_T}]$ , they show that  $y_\lambda(t) \leq At^{-\alpha}$  for all  $\lambda > 0$  and  $t > 0$ , obtained by combining a fractional maximum principle and that  $At^{-\alpha}$  is a strict supersolution of the VIE. This yields a lower bound for  $\mathbb{E}[e^{-\lambda V_T}]$  (uniformly in  $\lambda$ ) and letting  $\lambda \rightarrow \infty$  yields that  $\mathbb{P}(V_T = 0) > 0$ . The result shows that, unlike the classical CIR/Heston case or more general Volterra dynamics driven by nonsingular kernels discussed in [BP26] (see, in particular, Section 3.1 in [BP26]), no Feller-type inequality can make the zero boundary inaccessible for  $H < \frac{1}{2}$ . The same result is essentially shown in [FGW26], and in Section 1.5 they prove the surprising result that an rHeston model with constant drift  $\mu \neq r$  under  $\mathbb{P}$  admits no equivalent local martingale measure, and that equivalent martingale measures exist only under very restrictive assumptions on the drift under  $\mathbb{P}$ . Indeed, if  $L_t = \int_0^t \mathbf{1}_{\{V_s = 0\}} ds$ , then  $\mathbb{E}[L_t] = \int_0^t \mathbb{P}(V_s = 0) ds > 0$ , so  $\mathbb{P}(L_t > 0) > 0$ . Moreover, since  $V$  is continuous, the zero set  $\mathcal{Z} := \{s \in [0, t] : V_s = 0\}$  is closed, and  $L_t$  is precisely its Lebesgue measure. However, one can also easily show that  $V$  cannot be identically zero on any non-zero interval (see also Lemma 4.9 in [JP22]), so  $\mathcal{Z}$  contains no interval and hence has an empty interior. Thus,  $\mathbb{P}(\mathcal{Z} \text{ is closed, has positive Lebesgue measure, and has empty interior}) > 0$ , i.e., the zero set exhibits fat-Cantor-like behaviour with positive probability. In Theorem 1.5 in [FGW26], they also derive an exact closed-form expression for  $\mathbb{P}(V_T = 0)$  when the mean reversion is zero.

[JR20],[MORS26] model *signed order flow*  $F_t$  with Hawkes/Rough Heston-type processes as discussed above, and assume the price process is *endogenously* determined as  $P_t = \kappa \lim_{u \rightarrow \infty} \mathbb{E}[F_u | \mathcal{F}_t]$  (\*). If we assume instead that  $P$  is a given continuous semimartingale, and  $F_t = \int_0^t f(t-s) dP_s$ , with  $f(t) = ct^\gamma$  for  $c > 0$ ,  $\gamma \in (0, \frac{1}{2})$ , then we have the *inversion formula*:  $P_t = P_0 + \frac{d}{dt} \int_0^t h(t-s) F_s ds$  where  $h * f' \equiv 1$  (which is satisfied if  $h(t) = \frac{\sin(\pi\gamma)}{c\gamma\pi} t^{-\gamma}$ ). If  $P$  is also  $\frac{1}{2} - \varepsilon$ -Hölder-continuous for all  $\varepsilon > 0$ , by the standard mapping property of the operator  $F = cI^\gamma(\gamma+1)I^\gamma(P - P_0) \in C^{\gamma + \frac{1}{2} - \varepsilon}(0, T)$ <sup>1</sup>, so  $F$  is smoother than Brownian motion (consistent with empirical findings in [MORS26]). Since  $\gamma + \frac{1}{2} - \varepsilon > \gamma$  for  $\varepsilon \ll 1$ , Young integration by parts and the standard parameter-differentiation rule for Young convolutions imply that  $P$  can be rewritten as the Young integral  $P_t - P_0 = \int_0^t h(t-s) dF_s := \lim_{\delta \downarrow 0} \int_0^{t-\delta} h(t-s) dF_s$  (i.e. the *propagator equation*), which leads to *concave price impact* and the empirically well-documented square-root impact law in the limit as  $\gamma \uparrow \frac{1}{2}$  for additional exogenous trades with constant trading speed. This approach does not require the strong assumption (\*) above, or that  $P$  is driven by a hyper-rough Heston-type process as in [MORS26], and allows us to choose  $H$  and  $\gamma$  independently if  $P$  follows a rough volatility model with Hurst parameter  $H$ .

The mathematical finance literature also includes models with rough non-affine dynamics, e.g. the quadratic rough Heston (qRHeston) model in [GJR20], which has generally been more successful than the rHeston model in jointly fitting SPX and VIX smiles.<sup>2</sup> However, it is unclear whether the price process  $S$  for the model is a true martingale, since the drift for the  $Z$  process explodes quadratically when  $Z$  is large under the share measure  $\mathbb{P}^*$ , and for  $H = \frac{1}{2}$ , one can easily verify that  $S$  is a strict local martingale via the usual Feller boundary classification

<sup>1</sup>where  $I^\gamma$  denotes the  $\gamma$ -th order fractional integral.

<sup>2</sup>see e.g. [BG25] and notes/code on the second author's website.

for  $V$  under  $\mathbb{P}^*$ <sup>3</sup>. To circumvent this, we can just modify the drift or volatility coefficient of the SVE for  $Z$  (but that prevents us from using the exact VIX sampling formula in Section 6.2 in [Rom22]), or we can use e.g. the alternate generalized CEV Volterra model from Section 6.3 in [Rom22], for which this is not an issue. The price process  $S$  will of course be a martingale for a standard log-Euler Monte Carlo scheme.

This paper is organized as follows. In Section 2, we consider a rescaled version of (1)-(2) in the case  $\nu = 0$ , corresponding to the rough Heston model, and prove the convergence of the log-stock price marginals to those of an NIG process, see Theorem 2.1. In Section 3, we extend this approach to the jump setting  $\nu \neq 0$ , corresponding to the rough Hawkes–Heston model studied in [BLP24]. More precisely, Proposition 3.1 establishes the limiting behavior of the mgf of a suitable rescaling of the log-price process in (1). Theorem 3.2 in Subsection 3.1 then shows that the limiting expression corresponds to the mgf of a drifted time-changed Lévy process belonging to a class that extends the NIG family. The convergence results in Theorems 2.1 and 3.2 are numerically corroborated in Figures 1 and 3, and Figure 4 gives a sensitivity analysis for the asymptotic model in Theorem 3.2 with respect to the jump parameters  $\Lambda$  and  $\nu$ .

## 2 Asymptotics for the log stock price for a re-scaled rough Heston model

We consider processes defined on (possibly different) probability spaces  $(\Omega, \mathcal{F}, \mathbb{Q})$ , each equipped with a filtration  $(\mathcal{F}_t)_{t \geq 0}$  satisfying the usual conditions. We start by recalling the definition of Normal Inverse Gaussian distribution, which appears in the scaling limit in distribution established in Theorem 2.1.

**Definition 2.1** A random variable  $X$  is said to have a *Normal Inverse Gaussian (NIG)* distribution with parameters  $\hat{\alpha}, \delta > 0$ ,  $\mu \in \mathbb{R}$  and  $\beta \in (-\hat{\alpha}, \hat{\alpha})$ , denoted by  $X \sim NIG(\hat{\alpha}, \beta, \delta, \mu)$ , if its probability density function is

$$f_X(x) = \frac{\hat{\alpha}\delta}{\pi} \frac{K_1(\hat{\alpha}\sqrt{\delta^2 + (x - \mu)^2})}{\sqrt{\delta^2 + (x - \mu)^2}} \exp\left\{\delta\sqrt{\hat{\alpha}^2 - \beta^2} + \beta(x - \mu)\right\}, \quad x \in \mathbb{R},$$

where  $K_1$  denotes the modified Bessel function of the second kind.

The moment generating function of  $X \sim NIG(\hat{\alpha}, \beta, \delta, \mu)$  is given by

$$\begin{aligned} \mathbb{E}[e^{pX}] &= \exp\left\{\mu p + \delta\gamma - \delta\sqrt{\gamma^2 - p(p + 2\beta)}\right\}, \quad p \in (-\hat{\alpha} - \beta, \hat{\alpha} - \beta), \\ \text{where } \gamma &= \sqrt{\hat{\alpha}^2 - \beta^2}. \end{aligned} \tag{3}$$

An *NIG* process with parameters  $(\hat{\alpha}, \beta, \delta, \mu)$  is a (càdlàg) Lévy process  $L = (L_t)_{t \geq 0}$  where  $L_1 \sim NIG(\hat{\alpha}, \beta, \delta, \mu)$ . Such a process exists and is unique in law by Theorem 7.10 in [Sato99], which can be applied because (3) implies that the distribution  $NIG(\hat{\alpha}, \beta, \delta, \mu)$  is infinitely divisible. By, for instance, Example 1.3.32 in [Appl09], an *NIG* process  $L = (L_t)_{t \geq 0}$  can be constructed with a subordination approach as follows, recalling  $\gamma = \sqrt{\hat{\alpha}^2 - \beta^2}$  defined in (3):

$$L_t = \mu t + \beta \tau_t + W_{\tau_t}^{(1)}, \quad \text{with } \tau_t = \inf\{s \geq 0 : \gamma s + W_s^{(2)} = \delta t\}, \tag{4}$$

where  $W^{(1)}$  and  $W^{(2)}$  are two independent standard Brownian motions.

In the next theorem, Theorem 2.1, we consider a rescaled version of the dynamics in (1)-(2) with  $\nu \equiv 0$  (that is, the rough Heston model), and prove that the time- $t$  marginals of the log-price process converge to those of an *NIG* Lévy process independent of the initial variance  $V_0$  and the exponent  $\alpha$  in the fractional kernel. The proof relies on the affine structure of the model, which yields a semi-closed expression for the moment generating function of log prices in terms of deterministic Riccati–Volterra equations. In Figure 1, we visualize the weak convergence result in Theorem 2.1 at the level of implied volatility (IV) smiles.

**Theorem 2.1** Consider a rescaled rough Heston model for a log stock price process  $X_t^\varepsilon$ :

$$dX_t^\varepsilon = -\frac{1}{2}V_t^\varepsilon dt + \sqrt{V_t^\varepsilon}dB_t, \quad X_0^\varepsilon = 0, \tag{5}$$

$$V_t^\varepsilon = V_0 + \frac{1}{\varepsilon\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} (\lambda(\theta - V_s^\varepsilon)ds + \sigma\sqrt{V_s^\varepsilon}dW_s), \quad V_0 \geq 0, \tag{6}$$

where  $B$  and  $W$  are two Brownian motions with correlation  $\langle B, W \rangle_t = \rho t$ ,  $\rho \in (-1, 0]$ ,  $\alpha \in (\frac{1}{2}, 1)$  and  $\lambda, \theta, \sigma > 0$ . Denote by  $X = (X_t)_{t \geq 0}$  an *NIG* process with parameters

$$\beta = \frac{1}{\rho^2} \left( \frac{\lambda\rho}{\sigma} - \frac{1}{2} \right), \quad \hat{\alpha} = \sqrt{\beta^2 + \gamma^2}, \quad \text{with } \gamma = \frac{\lambda}{\sigma\rho}, \quad \delta = \lambda\theta\frac{\bar{\rho}}{\sigma}, \quad \mu = -\lambda\theta\frac{\rho}{\sigma}, \tag{7}$$

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<sup>3</sup>proof available on request

where  $\bar{\rho} = \sqrt{1 - \rho^2} > 0$ . Then, for all  $t > 0$  fixed,  $X_t^\varepsilon$  tends weakly to  $X_t$  as  $\varepsilon \rightarrow 0$ .

**Remark 2.1** As already mentioned before Theorem 2.1, the limiting NIG process  $X$  does not depend on  $V_0$  or the Hurst index  $H = \alpha - \frac{1}{2}$ , see the parameters in (7).

**Proof.** Denote by  $I^\alpha(f)(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} f(s) ds$  the  $\alpha$ -th order fractional integral of a function  $f$ . By, e.g., Section 4 in [ER19] or Section 7 in [ALP19], for  $p \in (0, 1)$

$$\mathbb{E}[e^{pX_t^\varepsilon}] = \exp \left\{ V_0 I^{1-\alpha} \phi_\varepsilon(t) + \frac{1}{\varepsilon} \lambda \theta I^1 \phi_\varepsilon(t) \right\}, \quad (8)$$

where  $\phi_\varepsilon$  is the unique continuous  $\mathbb{R}_-$ -valued solution of the Riccati–Volterra integral equation (VIE):

$$\phi_\varepsilon(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} \left( \frac{1}{2}(p^2 - p) + \frac{1}{\varepsilon}(\rho p \sigma - \lambda) \phi_\varepsilon(s) + \frac{1}{\varepsilon^2} \frac{1}{2} \sigma^2 \phi_\varepsilon(s)^2 \right) ds, \quad t \geq 0.$$

Now define the map  $\psi_q$  by  $\psi_q(t) = \frac{1}{\varepsilon} \phi_\varepsilon(\frac{t}{\varepsilon^q})$ , for some  $q \in \mathbb{R}$ . Then, denoting by

$$F(w) = \frac{1}{2}(p^2 - p) + (\rho p \sigma - \lambda)w + \frac{1}{2}\sigma^2 w^2, \quad w \in \mathbb{R},$$

the change of variables  $\varepsilon^q s = u$  shows that

$$\begin{aligned} \varepsilon \psi_q(t) &= \phi_\varepsilon\left(\frac{t}{\varepsilon^q}\right) = \frac{1}{\Gamma(\alpha)} \int_0^{t/\varepsilon^q} \left(\frac{t}{\varepsilon^q} - s\right)^{\alpha-1} \left(\frac{1}{2}(p^2 - p) + (\rho p \sigma - \lambda) \psi_q(\varepsilon^q s) + \frac{1}{2}\sigma^2 \psi_q(\varepsilon^q s)^2\right) ds \\ &= \frac{\varepsilon^{-q\alpha}}{\Gamma(\alpha)} \int_0^t (t-u)^{\alpha-1} F(\psi_q(u)) du. \end{aligned} \quad (9)$$

If we let  $q = -\frac{1}{\alpha}$ , the VIE (9) is independent of  $\varepsilon$ , and so is its unique continuous solution  $\psi = \psi_{-\alpha-1}$ . An application of Lemma 4.5 in [FGS21] (see also the proof of Theorem 2.1 in [BF25]) entails that

$$\lim_{t \rightarrow \infty} \psi(t) = U_1(p) := \frac{1}{\sigma^2} \left[ \lambda - p \sigma \rho - \sqrt{\lambda^2 - 2\lambda \rho \sigma p + \sigma^2 p(1 - p \bar{\rho}^2)} \right],$$

where  $U_1(p)$  is the smallest (equivalently, the negative) root of  $F$ , and we recall that  $\bar{\rho} = \sqrt{1 - \rho^2} > 0$ . Since, by definition,  $\phi_\varepsilon(t) = \varepsilon \psi(\varepsilon^{-1/\alpha} t)$ , for every  $t > 0$

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \phi_\varepsilon(t) = \lim_{s \rightarrow \infty} \psi(s) = U_1(p).$$

As a result, the nonpositive continuous function  $\frac{1}{\varepsilon} \phi_\varepsilon(\cdot)$  is bounded on  $\mathbb{R}_+$ , uniformly in  $\varepsilon$ . Therefore, for the exponent in (8), by the dominated convergence theorem

$$V_0 I^{1-\alpha} \phi_\varepsilon(t) + \frac{\lambda \theta}{\varepsilon} I^1 \phi_\varepsilon(t) = \frac{V_0}{\Gamma(1-\alpha)} \int_0^t (t-s)^{-\alpha} \phi_\varepsilon(s) ds + \lambda \theta \int_0^t \frac{1}{\varepsilon} \phi_\varepsilon(s) ds \xrightarrow{\varepsilon \rightarrow 0} 0 + \lambda \theta U_1(p) t. \quad (10)$$

By (3), explicit computations show that  $\lambda \theta U_1(p) t$  is the log moment generating function of an NIG Lévy process at time  $t$  with parameters  $(\hat{\alpha}, \beta, \delta, \mu)$  specified as in (7). Then, since  $p$  varies in  $(0, 1)$ , from, e.g., Problem 30.4 on Page 573 in [Bill86],  $X_t^\varepsilon$  tends weakly to the time- $t$  marginal law of an NIG( $\hat{\alpha}, \beta, \delta, \mu$ ) process, completing the proof. ■

### 3 Adding jumps: The Rough Hawkes–Heston model

We now generalize the processes  $(X^\varepsilon, V^\varepsilon)$  in (5)–(6) by introducing jumps. In particular, choosing again  $\alpha \in (\frac{1}{2}, 1)$ , for any  $\varepsilon > 0$  we consider

$$V_t^\varepsilon = V_0 + \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} \left( \frac{1}{\varepsilon} \lambda (\theta - V_s^\varepsilon) ds + \frac{1}{\varepsilon} \sigma \sqrt{V_s^\varepsilon} dW_s + \frac{1}{\varepsilon} d\tilde{J}_s^\varepsilon \right), \quad (11)$$

where  $\tilde{J}_t^\varepsilon = \int_0^t \int_{\mathbb{R}_+} x(N^\varepsilon(dx, ds) - V_s^\varepsilon \nu(dx) ds)$  and  $N^\varepsilon(dx, dt)$  is an integer-valued random measure with compensator  $V_t^\varepsilon \nu(dx) dt$ . As in [AJ21, BF25, BPS24], see also [ACLP21] and [BLP24], we assume  $\nu$  only has positive support with  $\nu(\{0\}) = 0$  and  $\int_{\mathbb{R}_+} |x|^2 \nu(dx) < \infty$ . Consequently,  $\tilde{J}^\varepsilon$  has positive-only jumps. Setting  $z = x/\varepsilon$  so  $dx = \varepsilon \cdot dz$ , we see that

$$\frac{1}{\varepsilon} \tilde{J}_t^\varepsilon = \frac{1}{\varepsilon} \int_0^t \int_0^\infty x(N^\varepsilon(dx, ds) - V_s^\varepsilon \nu(dx) ds) = \int_0^t \int_0^\infty z(N^\varepsilon(\varepsilon \cdot dz, ds) - V_s^\varepsilon \nu(\varepsilon \cdot dz) ds).$$

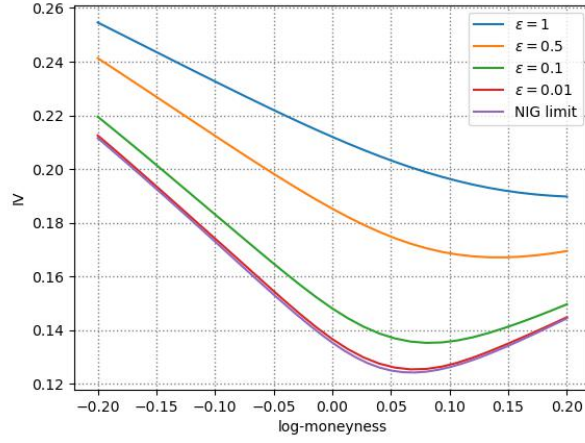


Figure 1: IV smiles for European options with underlying price processes  $e^{X_t^\varepsilon}$  and  $e^{X_t}$  defined in Theorem 2.1. Parameters:  $S_0 = 1$ ,  $\alpha = 0.7$ ,  $\rho = -0.5$ ,  $V_0 = 0.07$ ,  $\sigma = \frac{1}{2}$ ,  $\lambda = 1$ ,  $\theta = 0.04$  and maturity  $T = 1$ . Option prices are computed using the standard Lewis Fourier inversion formula (see, e.g., Eq. (1.6) in [GR24]). As  $\varepsilon \rightarrow 0$ , the convergence of the IV smiles to the limiting *NIG* smile becomes apparent.

As for the log stock price process  $X^\varepsilon = (X_t^\varepsilon)_{t \geq 0}$ , we model it with the dynamics

$$\begin{aligned} dX_t^\varepsilon &= -\left(\frac{1}{2} + \int_{\mathbb{R}_+} (e^{-\Lambda z} - 1 + \Lambda z) \nu(dz)\right) V_t^\varepsilon dt + \sqrt{V_t^\varepsilon} dB_t - \Lambda d\tilde{J}_t^\varepsilon \\ &= -\left(\frac{1}{2} + \int_{\mathbb{R}_+} (e^{-\varepsilon \Lambda z} - 1 + \varepsilon \Lambda z) \nu(\varepsilon \cdot dz)\right) V_t^\varepsilon dt + \sqrt{V_t^\varepsilon} dB_t - \varepsilon \Lambda \int_{\mathbb{R}_+} z \left(N^\varepsilon(\varepsilon \cdot dz, dt) - V_t^\varepsilon \nu(\varepsilon \cdot dz) dt\right), \end{aligned} \quad (12)$$

for a given leverage parameter  $\Lambda > 0$ . The couple  $(X^\varepsilon, V^\varepsilon)$  in (11)-(12) constitutes a rescaled rough Hawkes–Heston model (see (1)-(2) in Section 1), introduced and applied to real market data in [BPS24]. Note that, in the absence of jumps (that is,  $\nu \equiv 0$ ), the previous dynamics are continuous and reduce to (5)-(6).

The model in (11)-(12) is affine and [BPS24] provides a semi-closed formula for the moment generating function of  $X^\varepsilon$ . More specifically, define the map

$$V_\varepsilon(p, v) = \int_{\mathbb{R}_+} \left( e^{(v - \varepsilon \Lambda p)z} - p(e^{-\varepsilon \Lambda z} - 1) - 1 - vz \right) \nu(\varepsilon \cdot dz), \quad p \in [0, 1], v \leq 0; \quad (13)$$

we note that  $v \leq 0$  and  $\Lambda p \geq 0$ , together with the square-integrability assumption on  $\nu$ , ensure the convergence of the above integral, hence the finiteness of  $V_\varepsilon(p, v)$ . Indeed, since  $e^x - 1 - x \leq x^2$  for  $x \leq 0$ ,

$$\begin{aligned} |V_\varepsilon(p, v)| &\leq \int_{\mathbb{R}_+} \left( e^{(v - \varepsilon \Lambda p)z} - 1 - (v - \varepsilon \Lambda p)z \right) \nu(\varepsilon \cdot dz) + p \int_{\mathbb{R}_+} \left( e^{-\varepsilon \Lambda z} - 1 + \varepsilon \Lambda z \right) \nu(\varepsilon \cdot dz) \\ &\leq \left( (v - \varepsilon \Lambda p)^2 + p \varepsilon^2 \Lambda^2 \right) \int_{\mathbb{R}_+} |z|^2 \nu(\varepsilon \cdot dz) < \infty, \quad p \in [0, 1], v \leq 0. \end{aligned}$$

We also observe that the change of variables  $x = \varepsilon z$  yields

$$V_\varepsilon(p, \varepsilon v) = V_1(p, v), \quad p \in [0, 1], v \leq 0, \quad (14)$$

where  $V_1(p, v)$  is given in (13) with  $\varepsilon = 1$ . Consider now the function

$$F_\varepsilon(p, v) = \frac{1}{2}(p^2 - p) + \frac{1}{\varepsilon}(p\rho\sigma - \lambda)v + \frac{1}{2} \frac{1}{\varepsilon^2} \sigma^2 v^2 + V_\varepsilon(p, v), \quad p \in [0, 1], v \leq 0. \quad (15)$$

Then, from Theorem 3 in [BPS24], the moment generating function of  $X_t^\varepsilon$  is

$$\mathbb{E} \left[ e^{pX_t^\varepsilon} \right] = \exp \left\{ V_0 I^{1-\alpha} \phi_\varepsilon(t) + \frac{1}{\varepsilon} \lambda \theta I^1 \phi_\varepsilon(t) \right\}, \quad p \in (0, 1), \quad t \geq 0, \quad (16)$$

where  $\phi_\varepsilon: \mathbb{R}_+ \rightarrow \mathbb{R}_-$  is the unique (nonpositive) continuous solution of the deterministic Riccati–Volterra equation

$$\phi_\varepsilon(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} F_\varepsilon(p, \phi_\varepsilon(s)) ds, \quad t \geq 0. \quad (17)$$

In the next proposition, we study the limit of (16) as  $\varepsilon \rightarrow 0$ . This result is crucial for describing the limit (in distribution) of the marginal laws of  $X^\varepsilon$  in Subsection 3.1.

**Proposition 3.1** For every  $p \in (0, 1)$ ,

$$\text{define } U_J(p) \text{ as the unique negative root of the map } F_1(p, \cdot) \text{ on } \mathbb{R}_-, \quad (18)$$

where  $F_1(p, \cdot)$  is given in (15) with  $\varepsilon = 1$ . Then, for every  $t > 0$ ,

$$\lim_{\varepsilon \rightarrow 0} \mathbb{E} \left[ e^{pX_t^\varepsilon} \right] = \exp \left\{ \lambda \theta U_J(p) t \right\}. \quad (19)$$

**Proof.** The argument is analogous to that in the proof of Theorem 2.1, suitably adapted to account for jumps. Fix  $p \in (0, 1)$ . Introducing the function  $\psi(t) = \psi_{-\alpha^{-1}}(t) = \frac{1}{\varepsilon} \phi_\varepsilon(\varepsilon^{1/\alpha} t)$ , by the definition in (15) and (17) we find that, similarly to (9),

$$\begin{aligned} \varepsilon \psi(t) &= \phi_\varepsilon(\varepsilon^{1/\alpha} t) \\ &= \frac{1}{\Gamma(\alpha)} \int_0^{\varepsilon^{1/\alpha} t} (\varepsilon^{1/\alpha} t - s)^{\alpha-1} \left( \frac{1}{2}(p^2 - p) + (\rho p \sigma - \lambda) \psi(\varepsilon^{-1/\alpha} s) + \frac{1}{2} \sigma^2 \psi(\varepsilon^{-1/\alpha} s)^2 + V_\varepsilon(p, \varepsilon \psi(\varepsilon^{-1/\alpha} s)) \right) ds \\ &= \frac{\varepsilon}{\Gamma(\alpha)} \int_0^t (t - u)^{\alpha-1} \left( \frac{1}{2}(p^2 - p) + (\rho p \sigma - \lambda) \psi(u) + \frac{1}{2} \sigma^2 \psi(u)^2 + V_\varepsilon(p, \varepsilon \psi(u)) \right) du, \end{aligned}$$

where in the third line we set  $u = \varepsilon^{-1/\alpha} s$ . But since  $V_\varepsilon(p, \varepsilon \psi(\cdot)) = V_1(p, \psi(\cdot))$  (by (14)), the previous VIE is independent of  $\varepsilon$  and, by (15) with  $\varepsilon = 1$ , we can write

$$\psi(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t - u)^{\alpha-1} F_1(p, \psi(u)) du, \quad t \geq 0.$$

In order to study the asymptotic behavior of  $\psi$  using Lemma 4.5 in [FGS21], we now verify that  $F_1(p, \cdot)$  is (strictly) decreasing and analytic on  $\mathbb{R}_-$ . To this end, it is sufficient to show that  $V_1(p, \cdot)$  is nonincreasing and analytic on the nonpositive half-line. By its definition in (13) with  $\varepsilon = 1$ , the map  $V_1(p, \cdot)$  on  $\mathbb{R}_-$  can be decomposed as follows:

$$\begin{aligned} V_1(p, v) &= \int_{\mathbb{R}_+} (e^{-\Lambda p z} - p(e^{-\Lambda z} - 1) - 1 + e^{-\Lambda p z}(e^{vz} - 1) - vz) \nu(dz) \\ &= \int_{\mathbb{R}_+} (e^{-\Lambda p z} - p(e^{-\Lambda z} - 1) - 1) \nu(dz) + \int_{\mathbb{R}_+} e^{-\Lambda p z} (e^{vz} - 1 - vz) \nu(dz) + v \int_{\mathbb{R}_+} z(e^{-\Lambda p z} - 1) \nu(dz), \quad (20) \end{aligned}$$

for every  $v \leq 0$ . This shows that  $V_1(p, \cdot)$  is nonincreasing on  $\mathbb{R}_-$ , as it is given by the integral in  $\nu(dz)$  of decreasing functions, namely  $v \mapsto e^{vz} - 1 - vz$ , for every  $z \geq 0$ , and a line with nonpositive slope. Moreover, recalling that  $\Lambda > 0$  and  $p \in (0, 1)$ , differentiating  $V_1(p, \cdot)$  in (20), we see that, for every  $v \leq 0$ ,

$$V_1'(p, v) = \int_{\mathbb{R}_+} z(e^{-\Lambda p z} - 1) \nu(dz) + \int_{\mathbb{R}_+} z e^{-\Lambda p z} (e^{vz} - 1) \nu(dz), \quad V_1^{(n)}(p, v) = \int_{\mathbb{R}_+} |z|^n e^{vz - \Lambda p z} \nu(dz), \quad n \geq 2.$$

Hence  $V_1(p, \cdot)$  is of class  $C^\infty$  on  $\mathbb{R}_-$ . Notice that, for every  $n \geq 2$ , the function  $z \mapsto |z|^{n-2} e^{-\Lambda p z}$  attains its maximum on  $\mathbb{R}_+$  at  $z = \frac{n-2}{\Lambda p}$ , so

$$V_1^{(n)}(p, v) = \int_{\mathbb{R}_+} |z|^n e^{vz - \Lambda p z} \nu(dz) \leq \left( \int_{\mathbb{R}_+} |z|^2 \nu(dz) \right) n^n (\Lambda p e)^{2-n}, \quad v \leq 0, \text{ for every } n \geq 2.$$

Considering also that the first derivative of  $V_1(p, \cdot)$  is locally bounded on  $\mathbb{R}_-$ , by Stirling's approximation of the factorial this condition is sufficient to guarantee the analyticity of  $V_1(p, \cdot)$  on  $\mathbb{R}_-$ .

Therefore, the function  $F_1(p, \cdot)$  is analytic and (strictly) decreasing on  $\mathbb{R}_-$ . Moreover, using the decomposition of  $V_1(p, \cdot)$  in (20),

$$F_1(p, 0) = \frac{1}{2}(p^2 - p) + V_1(p, 0) = \frac{1}{2}(p^2 - p) + \int_{\mathbb{R}_+} (e^{-\Lambda p z} - p(e^{-\Lambda z} - 1) - 1) \nu(dz) < 0,$$

where the last inequality holds by (30) below and  $p \in (0, 1)$ . Furthermore, by straightforward estimates on  $\mathbb{R}_-$  based on (15) and (20),

$$F_1(p, v) \geq F_1(p, 0) + (\rho p \sigma - \lambda) v \xrightarrow{v \rightarrow -\infty} \infty.$$

These properties imply that  $U_J(p)$  in (18) is well-defined. Hence another application of Lemma 4.5 in [FGS21] yields, for every fixed  $t > 0$ ,

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \phi_\varepsilon(t) = \lim_{\varepsilon \rightarrow 0} \psi(\varepsilon^{-1/\alpha} t) = \lim_{s \rightarrow \infty} \psi(s) = U_J(p);$$

we refer to Figure 2 for a visualization of this convergence result. At this point, by the dominated convergence theorem (cf. (10)), we can take the limit as  $\varepsilon \rightarrow 0$  in (16) to deduce (19), which completes the proof. ■

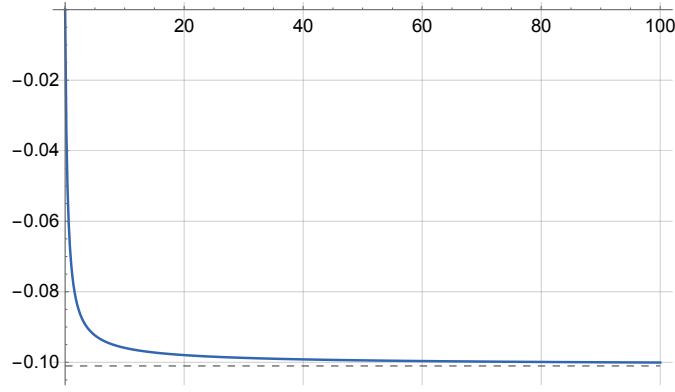


Figure 2: Here we see convergence of the VIE  $\psi(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-u)^{\alpha-1} F_1(p, \psi(u)) du$  to  $U_J(p)$  defined in (18) as  $t$  increases. This convergence result is crucial for the proof of Proposition 3.1. Parameters:  $\alpha = 0.7$ ,  $\sigma = 1$ ,  $\lambda = 1$ ,  $\rho = -0.5$ ,  $\Lambda = 1$ ,  $p = \frac{1}{2}$  and  $\nu(x) = \frac{C e^{-Mx}}{x^{1+Y}} 1_{x>0}$ , with  $C = 0.1$ ,  $M = 3$  and  $Y = 1.5$ . We use an Adams scheme with 500 time steps.

### 3.1 Characterizing the limiting law for $X^\varepsilon$ in terms of a subordinated Lévy process with drift

In this subsection, precisely in Theorem 3.2, we construct a time-changed Lévy process with drift  $X = (X_t)_{t \geq 0}$  whose time- $t$  marginal has log moment generating function  $\lambda \theta U_J(p)t$ , where  $U_J(p)$  is defined in (18). By Proposition 3.1, this implies that the log price  $X_t^\varepsilon$  in a rescaled rough Hawkes–Heston model converges to  $X_t$  in distribution as  $\varepsilon \rightarrow 0$ . In Remark 3.1, we argue that Theorem 3.2 reduces to Theorem 2.1 in Section 2 – where the distributional limit is an *NIG* process – when there are no jumps, i.e.,  $\nu \equiv 0$ . Finally, in Figure 3 we visualize the weak convergence result in Theorem 3.2 via IV smiles, and in Figure 4 we show the sensitivity of the IV smiles for the limiting model with log price process  $X$  with respect to the measure  $\nu$  and the leverage parameter  $\Lambda$ .

**Theorem 3.2** Define the spectrally positive Lévy process  $J = (J_t)_{t \geq 0}$  by

$$J_t = \int_0^t \int_{\mathbb{R}_+} z \left( N(dz, ds) - \frac{1}{\bar{\rho}^2} \nu(dz) ds \right), \quad t \geq 0,$$

where  $N(dz, dt)$  denotes a homogeneous Poisson random measure with intensity  $\frac{1}{\bar{\rho}^2} \nu(dz) \otimes dt$  and  $\bar{\rho} = \sqrt{1 - \rho^2} > 0$ . Consider two standard Brownian motions  $W = (W_t)_{t \geq 0}$  and  $\tilde{W} = (\tilde{W}_t)_{t \geq 0}$ , and suppose that  $J$ ,  $W$  and  $\tilde{W}$  are mutually independent. Set

$$\tilde{\beta} = \frac{1}{\bar{\rho}^2} \left( \frac{\lambda \rho}{\sigma} - \frac{1}{2} \right) - \frac{1}{\bar{\rho}^2} \int_{\mathbb{R}_+} (e^{-\Lambda z} - 1 + \Lambda z) \nu(dz), \quad \gamma = \frac{\lambda}{\sigma \bar{\rho}}, \quad \delta = \lambda \theta \frac{\bar{\rho}}{\sigma}, \quad \mu = -\lambda \theta \frac{\rho}{\sigma}. \quad (21)$$

Define the time-changed Lévy process with drift  $X = (X_t)_{t \geq 0}$  by

$$X_t = \mu t + \tilde{\beta} \tau_t + \tilde{W}_{\tau_t} - \left( \Lambda + \frac{\rho}{\sigma} \right) J_{\tau_t},$$

$$\text{where } \tau_t := \inf \left\{ s \geq 0 : \gamma s + W_s - \frac{\bar{\rho}}{\sigma} J_s > \delta t \right\}, \quad t \geq 0. \quad (22)$$

Then, for every  $t > 0$ ,  $X_t$  has log moment generating function  $t \lambda \theta U_J(\cdot)$  on  $(0, 1)$ , where  $U_J(p)$  is defined in (18). Consequently,  $X_t^\varepsilon$  defined in (12) converges weakly to  $X_t$  as  $\varepsilon \rightarrow 0$ .

**Remark 3.1** Observe that the parameters  $\gamma$ ,  $\delta$ , and  $\mu$  in (21) coincide with those in (7). In the no-jumps case, i.e., when  $\nu \equiv 0$ , the parameter  $\tilde{\beta}$  in (21) also coincides with  $\beta$  in (7). In particular, by (4), the process  $X$  defined in (22) is an *NIG*( $\hat{\alpha}, \beta, \delta, \mu$ ) Lévy process, where  $\hat{\alpha} = \sqrt{\beta^2 + \gamma^2}$ . Hence, in the continuous case  $\nu \equiv 0$ , Theorem 3.2 reduces to Theorem 2.1.

**Proof.** By, for instance, Theorem 25.17 in [Sato99],

$$\mathbb{E}[e^{v J_t}] = \exp \left\{ \frac{t}{\bar{\rho}^2} V_1(0, v) \right\}, \quad v \leq 0, t \geq 0, \quad (23)$$

where  $V_1$  is defined in (13) with  $\varepsilon = 1$ . Define the Lévy process  $\eta = (\eta_t)_{t \geq 0}$  by

$$\eta_t = \tilde{\beta} t + \tilde{W}_t - \left( \Lambda + \frac{\rho}{\sigma} \right) J_t, \quad t \geq 0.$$

We also introduce, for every  $p \in (0, 1)$  and  $u \leq 0$ , the process  $(\xi_t(p, u))_{t \geq 0}$  given by

$$\xi_t(p, u) = p \left( \eta_t + \frac{\mu}{\delta} \left( \gamma t + W_t - \frac{\bar{\rho}}{\sigma} J_t \right) \right) - \frac{u}{\delta} \left( \gamma t + W_t - \frac{\bar{\rho}}{\sigma} J_t \right), \quad t \geq 0. \quad (24)$$

Since  $W, \tilde{W}$  and  $J$  are mutually independent, using the moment generating function of the normal distribution we compute

$$\begin{aligned} \log \mathbb{E} \left[ \exp \left\{ \xi_t(p, u) \right\} \right] &= \left( p\tilde{\beta} + \frac{p\mu - u}{\delta} \gamma \right) t + \log \mathbb{E} \left[ e^{p\tilde{W}_t} \right] + \log \mathbb{E} \left[ \exp \left\{ \frac{p\mu - u}{\delta} W_t \right\} \right] + \log \mathbb{E} \left[ \exp \left\{ \left( \frac{\bar{\rho}}{\sigma\delta} u - \Lambda p \right) J_t \right\} \right] \\ &= \left( p\tilde{\beta} + \frac{p\mu - u}{\delta} \gamma \right) t + \frac{1}{2} \left( p^2 + \frac{1}{\delta^2} (p^2 \mu^2 + u^2 - 2\mu u p) \right) t + \frac{1}{\bar{\rho}^2} V_1 \left( 0, \frac{\bar{\rho}}{\sigma\delta} u - \Lambda p \right) t. \end{aligned}$$

Here, in the second line, to obtain the fourth term in  $J_t$ , we recall that  $\eta_t$  also depends on  $J_t$  and simplify two terms thanks to the relation  $\mu/\delta = -\rho/\bar{\rho}$ , which holds by (21). In the third line, we apply (23). Then, considering that, by the definitions of  $V_1(\cdot, \cdot)$  and  $\tilde{\beta}$  in (13) and (21), respectively,

$$\begin{aligned} p\tilde{\beta} + \frac{1}{\bar{\rho}^2} V_1 \left( 0, \frac{\bar{\rho}}{\sigma\delta} u - \Lambda p \right) &= \frac{1}{\bar{\rho}^2} \left[ p \left( \frac{\lambda\rho}{\sigma} - \frac{1}{2} \right) + \int_{\mathbb{R}_+} \left( e^{(\frac{\bar{\rho}}{\sigma\delta} u - \Lambda p)z} - p(e^{-\Lambda z} - 1 + \Lambda z) - 1 - \left( \frac{\bar{\rho}}{\sigma\delta} u - \Lambda p \right) z \right) \nu(dz) \right] \\ &= \frac{1}{\bar{\rho}^2} \left[ p \left( \frac{\lambda\rho}{\sigma} - \frac{1}{2} \right) + V_1 \left( p, \frac{\bar{\rho}}{\sigma\delta} u \right) \right], \end{aligned}$$

by algebraic manipulations based on (21), we complete the previous chain of equalities as follows:

$$\begin{aligned} \log \mathbb{E} \left[ \exp \left\{ \xi_t(p, u) \right\} \right] &= \frac{1}{\bar{\rho}^2} \left[ \frac{1}{2} (p^2 - p) + (\sigma\rho p - \lambda) \frac{u}{\lambda\theta} + \frac{1}{2} \sigma^2 \left( \frac{u}{\lambda\theta} \right)^2 + V_1 \left( p, \frac{u}{\lambda\theta} \right) \right] t. \end{aligned} \quad (25)$$

Recalling the map  $F_1(p, \cdot)$  defined in (15) with  $\varepsilon = 1$ , we then have

$$\log \mathbb{E} \left[ \exp \left\{ \xi_t(p, u) \right\} \right] = \frac{t}{\bar{\rho}^2} F_1 \left( p, \frac{u}{\lambda\theta} \right), \quad t \geq 0.$$

Choosing  $u = \lambda\theta U_J(p)$  with  $U_J(p)$  defined in (18), since  $(\xi_t(p, \lambda\theta U_J(p)))_{t \geq 0}$  has stationary and independent increments with respect to the natural filtration generated by the processes  $W, \tilde{W}$  and  $J$ , the previous equation implies that

$$\left( \exp \left\{ \xi_t(p, \lambda\theta U_J(p)) \right\} \right)_{t \geq 0} \text{ is a martingale.}$$

As in (22), for every  $t \geq 0$ , let  $\tau_t$  denote the first hitting time to the upper barrier  $\delta t$  of the (spectrally negative) Lévy process  $Y = (Y_s)_{s \geq 0}$  defined by  $Y_s = \gamma s + W_s - \frac{\bar{\rho}}{\sigma} J_s$ , namely

$$\tau_t := \inf \left\{ s \geq 0 : Y_s > \delta t \right\} = \inf \left\{ s \geq 0 : \gamma s + W_s - \frac{\bar{\rho}}{\sigma} J_s > \delta t \right\}.$$

Note that  $\mathbb{E}[Y_1] = \gamma = \frac{\lambda}{\sigma\bar{\rho}} > 0$ , so  $\tau_t < \infty$   $\mathbb{P}$ -a.s. (see Theorem 36.5 in [Sato99], and also Proposition C.1 in [BF25]). Then, by the optional stopping theorem and the definition of  $\xi_t(p, \lambda\theta U_J(p))$  in (24) we deduce that

$$\begin{aligned} \mathbb{E} \left[ \exp \left\{ \xi_{\tau_t \wedge n}(p, \lambda\theta U_J(p)) \right\} \right] &= \mathbb{E} \left[ \exp \left\{ p \left( \eta_{\tau_t} + \frac{\mu}{\delta} Y_{\tau_t} \right) - \lambda\theta \frac{U_J(p)}{\delta} Y_{\tau_t} \right\} 1_{\{\tau_t \leq n\}} \right] + \mathbb{E} \left[ \exp \left\{ p \left( \eta_n + \frac{\mu}{\delta} Y_n \right) - \lambda\theta \frac{U_J(p)}{\delta} Y_n \right\} 1_{\{\tau_t > n\}} \right] \\ &= 1, \quad n \in \mathbb{N}. \end{aligned} \quad (26)$$

Since  $\lim_{n \rightarrow \infty} 1_{\{\tau_t \leq n\}} = 1_{\{\tau_t < \infty\}} = 1$ ,  $\mathbb{P}$ -a.s., and  $Y_{\tau_t} = \delta t$  because  $Y$  is spectrally negative, from the monotone convergence theorem we see that the first term tends to

$$\mathbb{E} \left[ \exp \left\{ p(\mu t + \eta_{\tau_t}) - \lambda\theta U_J(p)t \right\} \right] = e^{-\lambda\theta U_J(p)t} \mathbb{E} \left[ e^{p(\mu t + \eta_{\tau_t})} \right] \quad \text{as } n \rightarrow \infty. \quad (27)$$

As for the second term, our goal is to use Vitali's convergence theorem to prove that

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[ \exp \left\{ p \left( \eta_n + \frac{\mu}{\delta} Y_n \right) - \lambda\theta \frac{U_J(p)}{\delta} Y_n \right\} 1_{\{\tau_t > n\}} \right] = 0. \quad (28)$$



To do this, we first observe that

$$\lim_{n \rightarrow \infty} \exp \left\{ p \left( \eta_n + \frac{\mu}{\delta} Y_n \right) - \lambda \theta \frac{U_J(p)}{\delta} Y_n \right\} 1_{\{\tau_t > n\}} = 0, \quad \mathbb{P} - \text{a.s.}, \quad (29)$$

again because  $\mathbb{P}(\tau_t = \infty) = 0$ . We then show that the sequence of random variables

$$\left( \exp \left\{ p \left( \eta_n + \frac{\mu}{\delta} Y_n \right) - \lambda \theta \frac{U_J(p)}{\delta} Y_n \right\} 1_{\{\tau_t > n\}} \right)_n \text{ is uniformly integrable.}$$

Since  $p \in (0, 1)$ , we can fix a  $q \in (1, \frac{1}{p})$ . Observing that, by the definition of  $\tau_t$ ,

$$\exp \left\{ -q \lambda \theta \frac{U_J(p)}{\delta} Y_n \right\} 1_{\{\tau_t > n\}} \leq e^{q \lambda \theta |U_J(p)|t},$$

we compute

$$\begin{aligned} \mathbb{E} \left[ \exp \left\{ q \left( p \left( \eta_n + \frac{\mu}{\delta} Y_n \right) - \lambda \theta \frac{U_J(p)}{\delta} Y_n \right) \right\} 1_{\{\tau_t > n\}} \right] &\leq e^{q \lambda \theta |U_J(p)|t} \mathbb{E} \left[ \exp \left\{ qp \left( \eta_n + \frac{\mu}{\delta} Y_n \right) \right\} \right] \\ &= e^{q \lambda \theta |U_J(p)|t} \exp \left\{ \frac{1}{2\rho^2} pq(pq - 1)n + \frac{1}{\rho^2} \left( \int_{\mathbb{R}_+} \left( e^{-\Lambda pqz} - pq(e^{-\Lambda z} - 1) - 1 \right) \nu(dz) \right) n \right\} \\ &\leq e^{q \lambda \theta |U_J(p)|t} \exp \left\{ \frac{1}{\rho^2} \left( \int_{\mathbb{R}_+} \left( e^{-\Lambda pqz} - pq(e^{-\Lambda z} - 1) - 1 \right) \nu(dz) \right) n \right\} \leq e^{q \lambda \theta |U_J(p)|t}, \quad n \in \mathbb{N}. \end{aligned}$$

Here, for the equality on the second line we use the moment generating function at 1 of  $qp(\eta_n + \frac{\mu}{\delta} Y_n) = \xi_n(pq, 0)$  (see (25) with  $u = 0$  and  $pq$  instead of  $p$ ), while for the inequalities on the third line we use, respectively,  $pq < 1$  and the elementary estimate

$$y^a \leq (1 - a) + ay, \quad y \geq 0, a \in (0, 1),$$

which, choosing  $y = e^{-\Lambda z}$  and  $a = pq$ , ensures that

$$e^{-\Lambda pqz} - pq(e^{-\Lambda z} - 1) - 1 \leq 0, \quad z \geq 0. \quad (30)$$

The previous computations show that

$$\sup_{n \in \mathbb{N}} \mathbb{E} \left[ \left| \exp \left\{ p \left( \eta_n + \frac{\mu}{\delta} Y_n \right) - \lambda \theta \frac{U_J(p)}{\delta} Y_n \right\} \right|^q 1_{\{\tau_t > n\}} \right] < \infty;$$

given that  $q > 1$ , this is a sufficient condition for the uniform integrability. Since (29) implies convergence in probability, we can apply Vitali's convergence theorem (see, e.g., Section 13.7 in [Will91]) to deduce (28). Combining (26), (27) and (28), we then conclude that

$$\mathbb{E} \left[ e^{p(\mu t + \eta_{\tau_t})} \right] = e^{\lambda \theta U_J(p)t}.$$

As  $p \in (0, 1)$ , recalling the definition of the process  $X = (X_t)_{t \geq 0}$  in (22), the previous equation shows that  $X_t = \mu t + \eta_{\tau_t}$  has log moment generating function  $t \lambda \theta U_J(\cdot)$  on  $(0, 1)$ . By Proposition 3.1, an application of Problem 30.4 on Page 573 in [Bill86] ensures that  $X_t^\varepsilon$  defined in (12) tends weakly to  $X_t$ , which completes the proof. ■

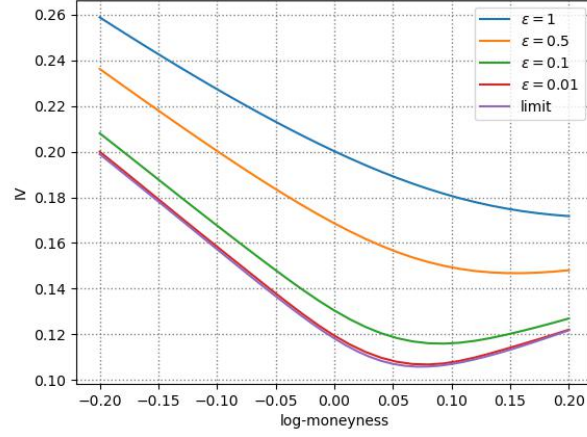


Figure 3: IV smiles for European options with underlying price processes  $e^{X_t^\varepsilon}$  and  $e^{X_t}$  defined in Theorem 3.2, using the same Fourier inversion formula and parameters as in Figure 1;  $\nu(dx) = e^{-x}dx$ ,  $\Lambda = 1$ . Even in this more general framework, as  $\varepsilon \rightarrow 0$ , the convergence of the IV smiles to the limiting smile with underlying  $e^X$  is apparent.

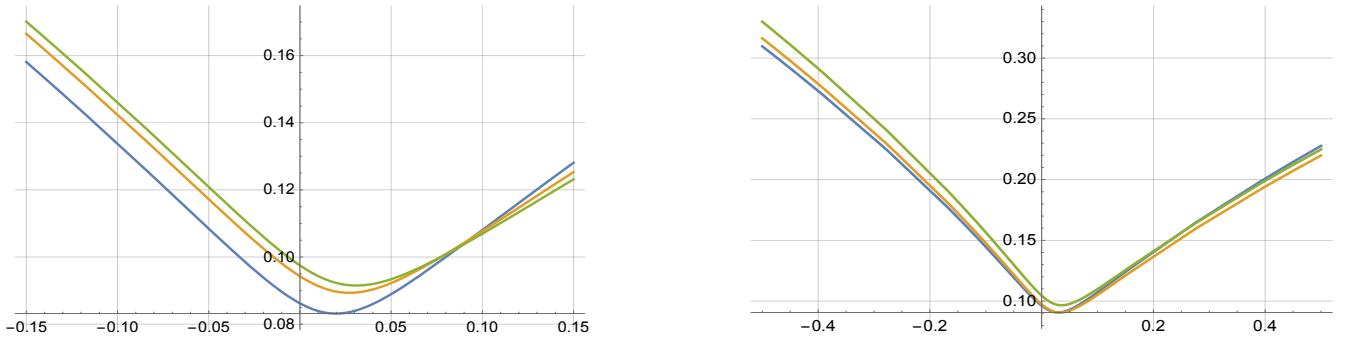


Figure 4: Implied volatility smiles as functions of log-moneyness for the asymptotic model  $S_t = e^{X_t}$  in Theorem 3.2, with  $\rho = -0.5$ ,  $\theta = 0.04$ ,  $\lambda = 1$ ,  $\sigma = 1$  and maturity  $T = 1$ . The measure  $\nu(dx)$  is the same as in Figure 2, but with  $M = 2$  and varying  $C$ . *Left:*  $C = 1$  (blue),  $C = 0.25$  (yellow) and  $C = 0.01$  (green), with  $\Lambda = 0$ . *Right:*  $C = 0.01$  fixed and  $\Lambda = 0, 1, 2$  (blue, yellow and green, respectively).

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